

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2025

Bachelor in Computer Applications

Course Title: **Cloud Computing**

Code No: **CACS402** Semester: **VII**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. Explain the key factors that organizations should consider before adopting cloud computing.
3. Explain the concept of Cloud design and implementation using Service-Oriented Architecture (SOA). What are the benefits of adopting SOA in Cloud computing?
4. Explain the benefits of virtualization in cloud computing. How does it improve resource utilization and reduce operational costs?
5. How does the Map-Reduce model work, and what are its advantages over traditional methods for processing large data sets?
6. Describe the key strategies for ensuring data security in a cloud environment?
7. Describe the role of cloud computing in business and consumer applications. How does cloud technology benefit businesses in terms of scalability and cost-effectiveness?
8. How do cloud computing platforms support scientific applications? Discuss the benefits of using the cloud for high-performance computing (HPC) in scientific research.

Group C

Attempt any TWO questions[2x10=20]

9. What is thread? Describe how thread programming is done in cloud. Discuss how virtualization is transforming the IT industry and its significance in cloud computing.
10. What are the critical aspects of application security in the cloud? Explain the techniques used to secure applications hosted in the cloud and prevent potential vulnerabilities from exploitation.
11. What is a Community Cloud? Discuss how it is different from Public and Private Clouds and explain the situations where Community Cloud deployment is beneficial.